



FUSION AI



SMART INDIA
HACKATHON
2024

AI-Driven Orthophoto Analysis for *SVAMITVA Scheme*

~Transforming Rural Land Records with Precision and AI



Problem [1705]

Development and Optimization of AI model for Feature identification/ Extraction from drone orthophotos. [\[1\]](#) [\[2\]](#)

Challenging Points:

- Lack of precise, scalable, and automated tools for **feature identification** from drone imagery.
- **Manual annotation** is time-intensive and prone to errors.
- **Inadequate resources** for applications like solar energy potential, property tax assessment, and infrastructure planning.

Impact:

- *Delayed decision-making* for rural development projects.
- *High costs* associated with manual processes.

Example Scenario:

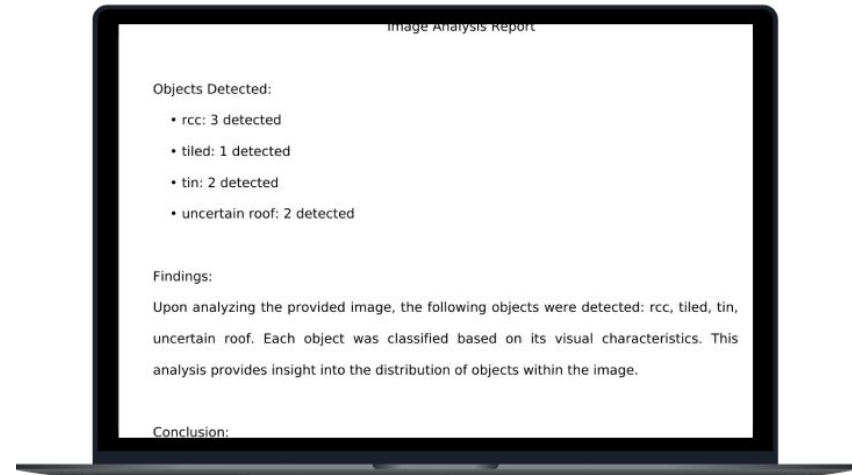
Inefficiencies in managing property disputes due to incomplete or inaccurate data.

Solution:

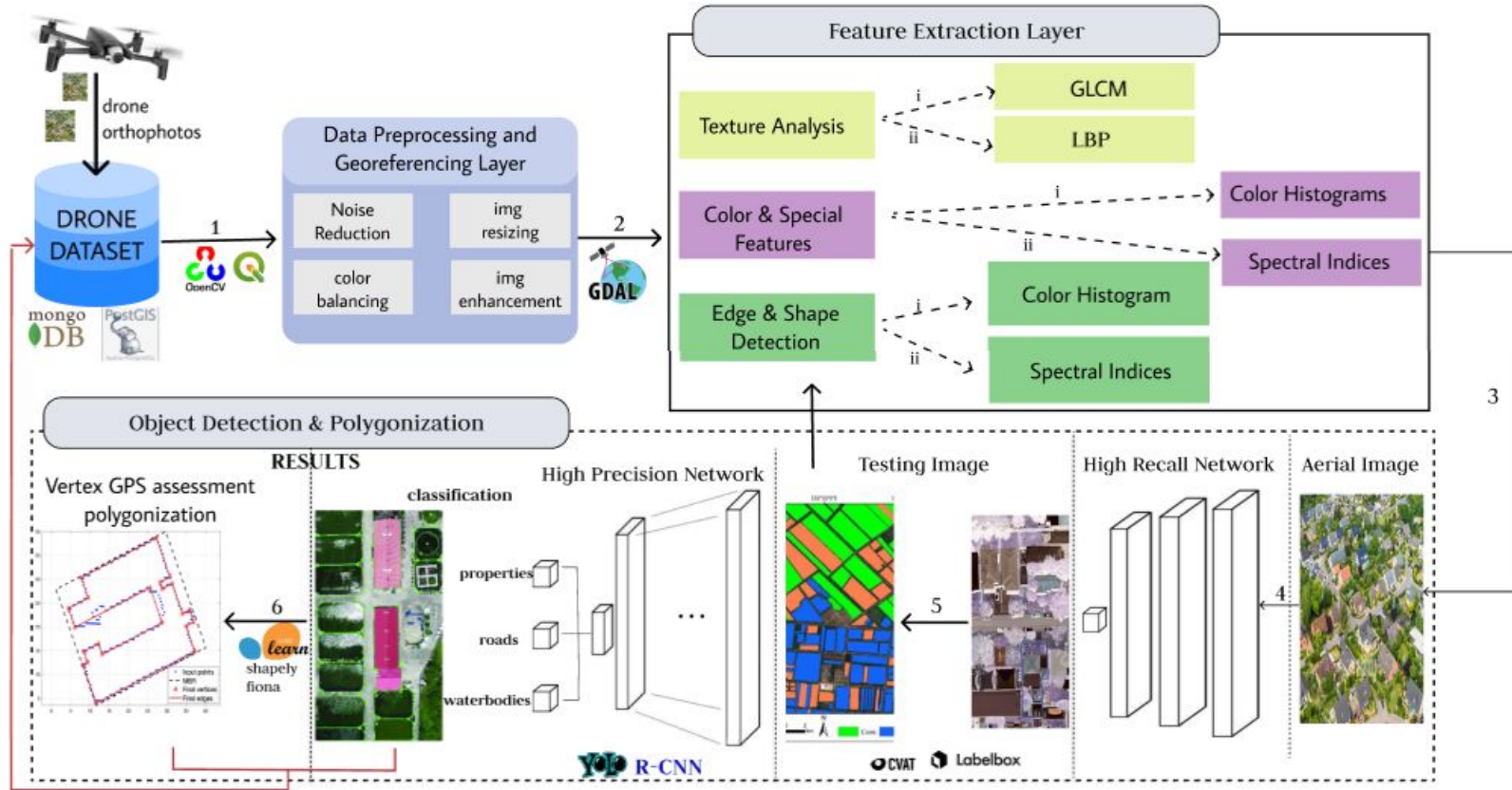
Paper reference: [Machine Learning Application in G.I.S. and Remote Sensing](#)

Automating Feature Extraction for Rural Transformation

- A **cloud-based AI model** that automates *feature identification* with *95% accuracy*.
- **Features Identified:**
 - Building footprints (*classified by rooftop type: RCC, Tiled, Tin, Others*).
 - Roads and pathways.
 - Water Bodies and other geographical elements.
- **Key Benefits:**
 - **Speed:** Processes images significantly faster than manual methods.
 - **Scalability:** Handles large datasets across multiple villages.
 - **Versatility:** Applications in solar energy, property tax, and dispute resolution.



AI Driven Model Architecture



Co-Ordinates [Flask]

FusionAI



English



Hi Deepanshu !

 Dashboard

 Upload

 Uploaded Images

 Predicted Images

 Reports & Analytics

 Find in Map

 Settings

 Log out

Upload an Image for Prediction

Choose Image

Choose File test101.jpg

 UPLOAD FROM GOOGLE

Latitude :

12.1111

Longitude (optional):

78.1119

Upload

Indirect Competitors

Esri (ArcGIS):



- A leading provider of GIS software with AI-powered mapping and analytics capabilities.
- Focuses on enterprise solutions with high cost and steep learning curves for non-experts.

DroneDeploy:



- Specializes in drone-based mapping for agriculture, construction, and property management.
- Less focus on village-level governance and lacks integration with Indian government systems.

Pix4D



- Provides tools for drone orthophoto processing and 3D modeling.
- Primarily serves urban infrastructure and construction sectors, not optimized for SVAMITVA's requirements.

Trimble:



- Provides geospatial and surveying tools, primarily hardware-focused.
- Requires third-party software for AI integration.

Competitive Advantages / UVP

[GIS] [2.3]



- Adapted for rural applications under the SVAMITVA Scheme.
- Higher accuracy (98% target) and speed compared to manual methods.
- Cost-effective and scalable cloud-based solution.
- Seamless integration with existing government systems.

Feature	FUSION AI	GIS Tools	Specialized Services
Accuracy	95% + precision for rural feature extraction	Standard accuracy, less precise	High accuracy, but expensive
Speed	Fast, automated processing	Slower, manual input required	Moderate speed, needs integration effort
Cost	Affordable, cloud-based solution	Expensive licensing and resources	High cost, not suitable for small-scale
Customization	designed for SVAMITVA Scheme use cases	General tools, not rural-specific	Limited customization for rural planning
Integration	Seamless API integration with Ministry	Complex setup for large datasets	Integration requires significant effort

Business Model Canvas

FUSION AI

Key Partners  - Ministry Of Panchayati Raj GIRS & CORS Technology Providers - AWS Cloud Deployment	Key Activities  - Annotating SWAMITVA drone imagery Achieving 98% accuracy - Integrating with existing govt data	Value Propositions  - Identification Of property boundaries & features. - Facilitation of Bank loans and property monetization - Resolving property dispute - Village level planning with GIS data.	Customer Relationships  Government: close collab thr workshops. Rural Household: transparent communication 'bout demarcation	Customer Segments  Primary: Ministry of Panchayati Raj (SVAMITVA Scheme). Secondary: Rural Property owners.
Cost Structure  - Free of cost for rural landowners and Govt - Advanced services, cost have to be paid for respective platforms			Revenue Streams  Payment for premium services only	

Target Market:

- **Primary:** Ministry of Panchayati Raj (SVAMITVA Scheme).
- **Secondary:** Rural Property owners
- **Tertiary** : Urban development authorities, private infrastructure developers, renewable energy firms.

Market Size [\[1\]](#) [\[2\]](#)

- Government initiatives for rural and urban planning in India.
- Renewable energy market growth (solar potential analysis).
- Increasing adoption of drones in mapping and survey.
- **Stats:**
 - ₹63,000 crores potential market for geospatial solutions in India, as the Indian geospatial economy is projected to reach this value by 2025, growing at a rate of 12.8% annually

IMPACT OF OUR SOLUTION [\[1\]](#) [\[2\]](#) [\[3\]](#)

Efficiency in Property Management:

- **Time Savings:** *Reduces manual survey time* by up to 80%, enabling faster resolution of property disputes.
- **Cost Savings:** Significantly *lowers the cost of surveys and record maintenance*, saving up to ₹15 lakh per village.
- **Improved Accuracy:** Achieves *over 90% precision* in feature identification, minimizing errors compared to manual methods.

Enhanced Governance:

- **Transparency:** Creates tamper-proof digital land records, improving trust in governance.
- **Policy Planning:** Enables village-level planning with accurate mapping of roads, buildings, and waterbodies.

Support for Renewable Energy:

- Enables detailed rooftop analysis for solar potential, aligning with India's renewable energy goals.
- Facilitates data-driven decisions in solar energy implementation.

Environmental Benefits:

- Assists in monitoring and managing water bodies, aiding in conservation efforts.
- Provides data for green growth initiatives like afforestation and waste management.

SCALING THE VISION

- **Short-term:**
 - Deployment in pilot regions (10 villages).
 - Integration with Ministry systems.
- **Mid-term:**
 - Expand to additional regions.
 - Add support for more feature types (e.g., vegetation).
- **Long-term:**
 - AI-as-a-Service for rural and urban governance globally.

TRACTION [\[1\]](#) [\[2\]](#) [\[3\]](#)

Achievements

1. **Model Accuracy:** Model trained and validated on **10 villages with 94% current accuracy**.
(Based on benchmarks for similar geospatial segmentation models reaching 90-95%)

Metrics

1. **Time Saved:** **7-8 hours saved** per hectare compared to manual surveys.
(Manual surveys take ~10 hours; drone-based surveys achieve this in ~2 hours) .
2. **Cost Savings:** **₹15 lakh saved** per village on average.
(Manual surveys cost ₹25-30 lakh per village; AI-powered drone surveys cost ₹10-15 lakh)

References

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